Attention-Based Multistage Fusion Network for Remote Sensing Image Pansharpening

Wanwan Zhang, Jinjiang Li, and Zhen Hua

***Abstract*— Pansharpening is a significant branch in the field of remote sensing image processing, the goal of which is to fuse panchromatic (PAN) and multispectral (MS) images through certain rules to generate high-resolution MS (HRMS) images. Therefore, how to improve the spatial and spectral resolutions of the fused image is the problem that we need to solve urgently. In this article, a multistage remote sensing image fusion network (MRFNet) is proposed on the basis of in-depth research and exploration on the fusion of the PAN and MS images to obtain a clear fused image that can reflect the ground features more comprehensively and completely. The proposed network consists of three stages that are connected by cross-stage fusion. The first two stages are used to extract the features of the PAN and MS images. The structure of the encoder–decoder and the channel attention module are used to extract the features of the remote sensing image in the channel domain. The third stage is the image reconstruction stage fusing the extracted features with the original image to improve the spatial and spectral resolutions of the fused result. A series of experiments are conducted on the benchmark datasets WorldView II, GF-2, and QuickBird. Qualitative analysis and quantitative comparison show the superiority of MRFNet in visual effects and the values of evaluation indicators.**

***Index Terms*— Attention, encoder–decoder, multispectral (MS) images, panchromatic (PAN) images, pansharpening, spatial resolution, spectral resolution.**

I. Introduction

REMOTE sensing technology can observe places through sensors that humans cannot reach in a short time, which serves all aspects of people's production and life, such as meteorological observation, target detection [1], and image segmentation. With the improvement of sensor technology, the image bands acquired by remote sensing platforms have evolved from a single band to multiple bands, making spectral information more abundant. Currently, common remote sensing images include panchromatic (PAN) images, multispectral (MS) images, and hyperspectral (HS) images, the spatial and spectral resolutions of which are all different.

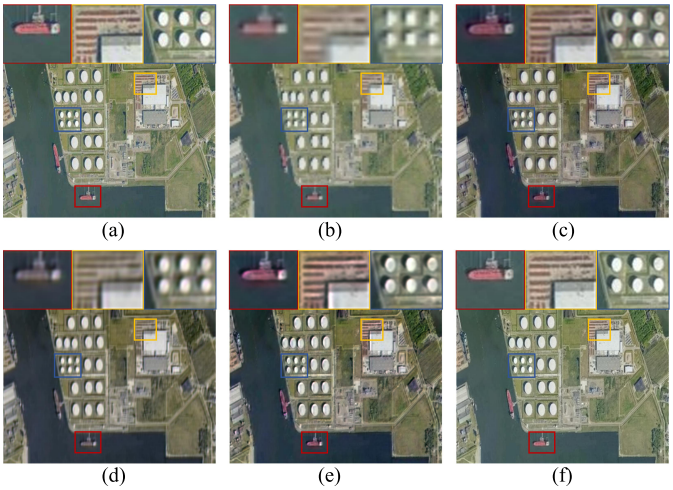

Fig. 1. Schematic illustration of pansharpening. (a) Reference (ground truth) and (b) MS images, respectively. Pansharpening results of (c) PanNet, (d) PNN+, (e) PanGAN, and (f) our proposed MRFNet, respectively.

Due to the limitations of existing sensors, the spectral and spatial resolutions of remote sensing images are mutually restricted. In other words, it is difficult for the remote sensing images that we obtain to have both high spatial resolution and high spectral resolution. Thus, how to obtain rich features from different remote sensing images and further improve the spatial and spectral resolutions has become a hot research topic in the field of remote sense (RS). Therefore, to obtain images with high spatial and spectral resolutions at the same time, the technology of pansharpening came into being. Pansharpening is an important branch in the field of remote sensing image fusion, the principle of which is to fuse the spatial information in the PAN image and the spectral information in the MS image to generate a high-resolution MS (HRMS) image rich in spatial information and spectral information (see Fig. 1).

The existing methods for pansharpening are mainly divided into three categories: component substitution (CS)-based methods [2]–[4], multiresolution analysis (MRA)-based methods, and model-based methods. The CS-based methods, including BROVEY transform [5], intensity–hue–saturation (IHS) [6], [7], principal component analysis (PCA) [8],

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Gram–Schmidt (GS) [9], [10], and the band, depend on the spatial details (BDSD) [11], [12]. More on the basis of the IHS and GS methods, Rahmani *et al.* [13] put forward AIHS, and Aiazzi *et al.* [2] proposed GS Adaptive (GSA). The CS-based methods for pansharpening retain the spatial details of the image better and are easy to implement, but it is easy to cause distortion of the spectral information. In order to improve this situation, the subsequent CS-based methods are mainly improved and deformed in the structure of the replacement component [14]–[16]. For example, the GFPCA proposed by Liao *et al.* [17] is an improvement on the traditional PCA method, which is a classic algorithm for applying a guided filter in the PCA domain.

In order to enhance spectral information, MRA-based methods [18] began to appear in people's field of vision. Commonly used MRA methods include à trous wavelet transform (ATWT) [19], [20], wavelet transform [21], Laplacian Pyramid (LP) [22], [23], high-pass filter (HPF) [24], and so on. However, the type of methods may introduce noise during the fusion process and makes spatial information loss. Considering that the CS-based method and the MRA-based method have their own advantages and disadvantages, many methods combine the two for remote sensing image pansharpening. For example, Zhong *et al.* [25] combined the CS-based method with the MRA-based method and proposed a new BDSD method for pansharpening; Wady *et al.* [26] combined the IHS-based method and ATWT to generate high-quality fused images.

Another type of model-based methods [27], [28] has also received widespread attention in recent years, which regards pansharpening as a local optimization problem. Currently, the typical methods are to use sparse representation for pansharpening. For the first time, Li and Yang [29] used sparse representation in the field of pansharpening of remote sensing images. However, the method was limited by the dictionary construction method. To solve this problem, Ayas *et al.* [30] proposed to generate a single dictionary from the texture information of the HRMS image to enhance the robustness. SparseFI based on compressed sensing theory was proposed by Zhu and Bamler [31], which used the sparse consistency of MS image and HRMS image patch for sparse fusion. On this basis, Fei *et al.* [32] proposed the pansharpening method based on injecting details and joint convolution SR (CSR) instead of the traditional SR model to overcome the shortcomings of incomplete spatial detail preservation. The results of the model-based method are better than the first two methods, but the complexity is higher, so the calculation time is longer.

With the unremitting efforts of scientific researchers, pansharpening technology based on deep learning has also achieved fruitful results. However, for the deep-learning-based methods for pansharpening, there are still two problems in the fused result: incomplete preservation of spatial information and distortion of spectral information due to its unreasonable network structure or improper parameter settings. To this end, we propose a multistage remote sensing image fusion network (MRFNet) for pansharpening. In this article, we recognize pansharpening as a multistage task and conduct experiments on three benchmark datasets to prove the superiority of our model. The main contributions of this work are given as follows.

1) We propose an effective pansharpening framework, which uses a multistage structure to reconstruct the HRMS image from coarse to fine and combines the channel attention module CAM) and the supervision attention module (SAM) to enable the network to learn important spatial information and spectral information and improve the network's ability to retain information. Finally, the HRMS image with complete spatial details and uniform spectral distribution is generated.
2) The first two stages of the network adopt the design of the encoder–decoder, which echoes the channel attention mechanism to complete the splicing and transmission of features in the channel domain. In the encoder, the inverted residual block is used instead of the conventional downsampled to enhance the expressive ability of the network and reduce the amount of calculation and increase the robustness of the network.
3) Except for the skip connection between encoder and decoder, the cross-stage fusion and SAM are added between every two stages, which makes the features extracted by the network more comprehensive. In addition, the resolution enhancement nets (RENets) are added in the third stage to further enable the fused result to cover rich spatial information and spectral information.

The rest of this article is organized as follows. In Section II, the pansharpening methods based on deep learning and background materials related to the work are introduced. In Section III, the proposed method and network structure are described in detail. In Section IV, the datasets, evaluation indicators, and analysis results related to the experiment are discussed. In Section V, a conclusion of this article is drawn.

II. Related Work

In this section, the relevant background materials on which our work is based, including the pansharpening method based on deep learning and the development of the attention mechanism, are introduced.

A. Deep-Learning-Based Methods for Pansharpening

The first application to the field of pansharpening was the super-resolution method [33]. Masi *et al.* [34] proposed to use PNN for PAN and MS images fusion. The network was simple and effective, but only had a three-layer network, so it was difficult to extract deep effective features. On this basis, Scarpa *et al.* [35] made an improvement and proposed PNN+. PanNet [36] and AMDF-ResNet [37] were proposed for the PAN and MS images pansharpening, which used ResNet as a deep model to extract and retain spatial information and spectral information. Wei *et al.* [38] proposed to introduce the concept of residual learning into pansharpening to design a deep network DRCNN, which expanded the number of layers of the network and improved the ability of the network to extract features. On this basis, MSDCNN [39] was proposed to achieve the pansharpening of remote sensing

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images. In addition to the introduction of residual learning, MSDCNN added multilevel feature extraction to extract as many spatial and spectral features as possible. Shao and Cai [40] designed a network structure with two branches, called RSIFNN, which can extract the spatial and spectral features of PAN and MS images and use the extracted features to generate a fusion image finally. Xu *et al.* [41] proposed the method of partially guided convolutional sparse coding with side information, called SCSC-PNN, which extended the model to deep learning. The remote sensing image fusion algorithm based on CoF and MSMG-PCNN was proposed, called CoF-MSMG-PCNN [42]. This method decomposed the PAN image into small-scale images, large-scale images, and basic images for fusion, but the fused results were prone to blurring. Liu *et al.* [14] proposed a two-stream fusion network (TFNet), which consisted of three parts: feature extraction, feature fusion, and image reconstruction. In the PSGAN [43], RED-cGAN [44] and PanGAN [45], which were based on generative adversarial networks, used the generator and discriminator to maintain the spatial information of PAN images and the spectral information of MS images.

B. Attention Mechanism

In the field of image processing, researchers have designed an attention mechanism based on space or channel based on the CNN model or the RNN model. Mnih *et al.* [46] introduced the attention mechanism into the RNN model to improve the accuracy of image classification and recognition. In addition, a residual attention network [47] was proposed for image classification, which was formed by stacking attention modules and combined an end-to-end training method with a feedforward network architecture. The squeeze and excitation networks (SENets) [48] proposed by Hu *et al.* designed the squeeze-and-excitation module on the basis of focusing on the relationship between channels and the importance of channel features, which introduced attention to the feature map channel mechanism. Recently, some attention-based models have also been proposed to further improve the performance of image processing. On the basis of deep residual channel network (RCAN) [49], the CAM was added to adaptively adjust the relationship between channels. The second-order attention network (SAN) was proposed for single image super-resolution by Dai *et al.* [50], who developed and added a second-order channel attention (SOCA) module, which adaptively adjusted the channel characteristics by using second-order feature statistics. Furthermore, an efficient channel attention (ECA) module [51] and a lightweight enhanced spatial attention (ESA) block [52] were designed to maintain performance, which can make the models more focused on the key channel and spatial content and give full play to the functions of the attention mechanism.

III. Proposed Method

In this section, we introduce the MRFNet proposed in this article for fusing PAN and MS images to generate HRMS images. First, we introduce the motivation of the work. Then, the overall framework and important modules of the network are described. Finally, we present the loss function used in the training phase.

A. Motivation

The remote sensing platform cannot directly obtain MS images with high spatial resolution due to the limitations of existing sensors. Generally available are PAN images with high spatial resolution and low spectral resolution and MS images with low spatial resolution and high spectral resolution. As mentioned above, the fused results generated by the CS-based methods have the problem of spatial details loss, while the fused results on the MRA-based methods have the problem of spectral distortion. Model-based methods are more complex, while methods based on deep learning can improve the spatial and spectral resolutions of the fused image significantly. The purpose of pansharpening is to obtain the MS image with the same spatial resolution as the PAN image, so the network that we design needs to get the fused result, which is closed to the ground truth.

The original PAN image contains rich spatial detail information, while the spectral information in the MS image is very significant. If the features are extracted separately from the image, it may increase the number of network calculations. Therefore, we consider using the coarse hierarchical structure of the image to reconstruct the fine hierarchical structure. This course-to-fine idea has been applied and achieved great results. For example, a multiscale convolutional neural network [53] is proposed to fulfill the image deblurring task, which is divided into the coarsest layer and the finer layer. In the network, the output information of the coarsest layer is connected with the fuzzy block of the finer scale, as the input of the network of the finer layer. At the finest layer, the original blurred image is restored to a clear image. Inspired by the use of the coarse hierarchical structure of the image to reconstruct the fine hierarchical structure, our network is divided into three stages. The fused result of the PAN and MS images is the input of the network, making full use of the advantages of the encoder–decoder and CAM to extract and restore the spatial and spectral resolutions of the input image layer by layer gradually.

B. Architecture of Proposed Network

Pansharpening of remote sensing image is to integrate the PAN and MS image according to certain rules to integrate spatial details and spectral information. On the basis of improving the spatial resolution of the fused result, the spectral information is maintained by the model as much as possible, which obtains the HRMS image finally. As depicted in Fig. 2, the whole network consists of three stages, among which cross-stage fusion is added between Stage 1 and Stage 2, and between Stage 2 and Stage 3 [the red dotted line in Fig. 2(a)]. Before Stage 1, the MS image is upsampled to the same size as the PAN image, and then, the two images are concatenated together to obtain the original fused image as the input of Stage 1.

Stage 1 and Stage 2 both use the structure of the encoder–decoder to extract features, which is shown

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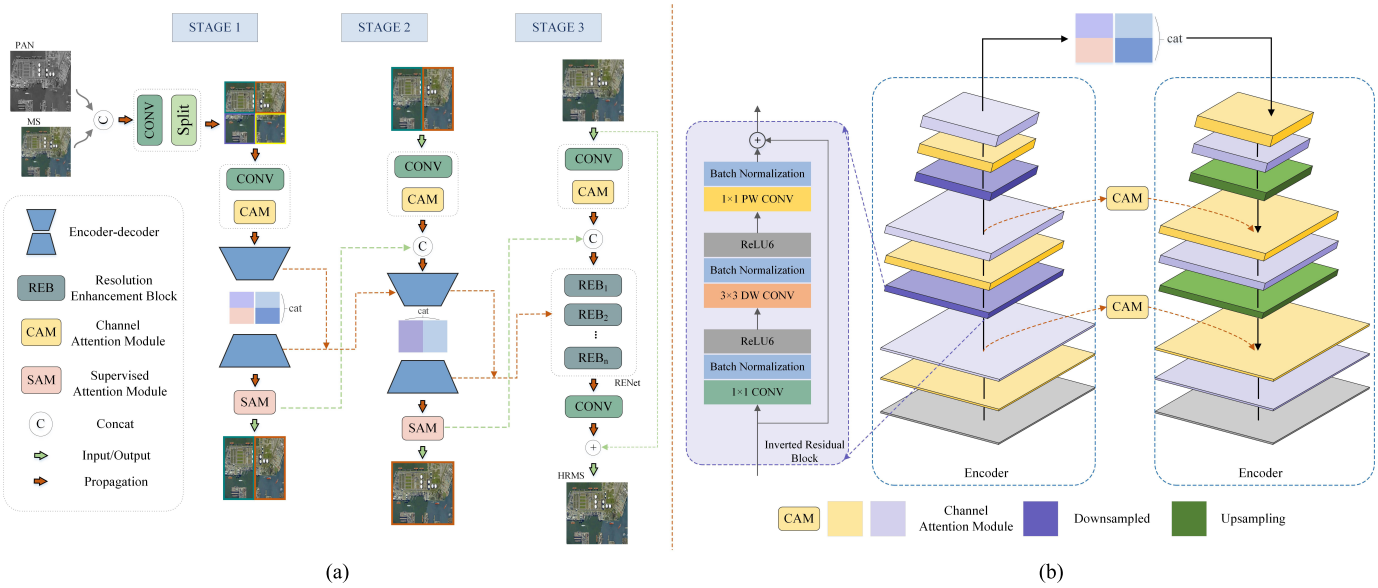

Fig. 2. Overall framework of MRFNet. The network consists of three stages: Stage 1, Stage 2, and Stage 3. (a) Structure of MRFNet. (b) Detailed structure of encoder–decoder in Stage 1 and Stage 2.

in Fig. 2(b). To extract precise spatial details and spectral information while reducing the amount of calculation as much as possible, we use the inverted residual block in MobileNetV2 as the basic structure of downsampled in the encoder. In addition, unlike the traditional skip connection, the feature map in the encoder needs to be connected to the corresponding feature map in the decoder via the CAM, which greatly improves the retention of spatial and spectral features. It is worth noting that the connection between the stages is not to directly transmit the feature map but to add an SAM between every two stages to give full play to the role of each stage, which realizes the spread from coarse to fine and between stages gradually.

Stage 3, as the image reconstruction stage, adopts a different structure from the previous two stages. The feature map in Stage 2 through cross-stage fusion enters the RENet, which is added to the original fusion image to obtain the fused result of this work finally.

1) Inverted Residual Block: In the above introduction, we learned that the inverted residual block in MobilenetV2 [54] is used to replace the traditional downsampled in the encoder of the first two stages. The detailed structure of the inverted residual block is shown in Fig. 2(b). The module can reduce the number of network calculations while extracting fine features for the use of depthwise (DW) separable convolutions, which consists of DW convolution and pointwise (PW) convolution. DW convolution uses different convolution kernels for each channel, and PW convolution is the ordinary convolution using 1*1 convolution kernels. In the inverted residual block, the feature map is first expanded by a 1*1 convolution kernel to expand the number of channels, where the hyperparameter of the dimension expansion is set to 6. Then, a 3*3 DW convolution is used to extract features (reducing the amount of calculation), and finally, PW convolution is used to map the high-dimensional space into the low-dimensional space. In this way, the downsampled feature map is completed.

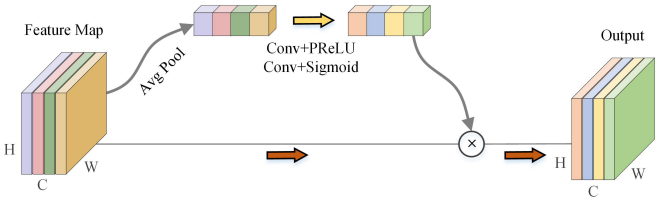

Fig. 3. Detailed structure of the CAM. W and H refer to the spatial dimension of the feature map, and refers to the number of channels of the feature map. The size of the output feature map is the same as the input feature map.

2) Channel Attention Module: Considering that different channels in the feature map are of different importance in feature extraction, the CAM is introduced to the network, which can assign different weights to different channels on the feature map to achieve selective learning. In the feature learning of this work, the attention mechanism can combine the channel attention method [48] to extract more effective spatial information and spectral information and suppress noise, which helps the network to learn the features of the PAN and MS image synthesis and improve the fusion ability of details in a feature map.

Fig. 3 shows the application of the CAM in this work. The feature map with the size of $W \times H$ and the number of C is used as the input of the CAM. The input is first compressed into a set of real numbers with a shape of $1 \times 1 \times C$ by global average pooling so that it has a global receptive field and the underlying network can also use global information. Then, the abovementioned set of real numbers is subjected to a nonlinear transformation through a fully connected neural network, which uses PReLU and sigmoid functions. The PReLU function is an improvement to the ReLU function, where learnable parameters are added to solve the possible necrosis problem of the ReLU function. The two activation functions are defined as follows:

$$\text{Sigmoid}(x_i) = \frac{1}{1 + e^{-x_i}} \tag{1}$$

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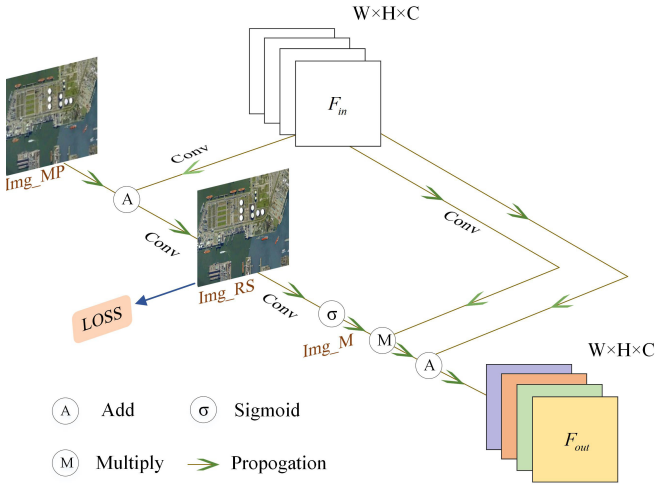

Fig. 4. Specific structure of the SAM. The input feature map is F_{in}, and the output feature map is F_{out}. Img_MP is the original fused image of the PAN and MS image. Img_RS is the fused result recovered at the stage of SAM, which is calculated the loss with ground truth.

$$\text{PReLU}(x_i) = \begin{cases} x_i, & \text{if } x_i > 0 \\ \alpha x_i, & \text{if } x_i \leq 0 \end{cases} \tag{2}$$

where α takes the number between [0, 1] generally.

In this step, the correlation between feature map channels can be explicitly modeled by generating weights for each channel through parameters. Therefore, the result of this step is also a set of real numbers representing the degree of attention of the feature map channel (see Fig. 3). Finally, the result obtained in the previous step is used as the weight, which represents the importance of the feature map channels. The set of weights is multiplied by the input feature map to complete the recalibration of the feature map in the channel dimension. Therefore, the resulting output by the CAM is a feature map recalibrated to the importance of the channel with the spatial dimension of $W \times H$ and the channels number of C.

3) Supervision Attention Module: In addition to adding CAM in each stage, we also add the SAM [55] between every two stages to improve the accuracy of the fusion. The flowchart of SAM is shown in Fig. 4, acting between the first and second stages and between the second and third stages. In the proposed method, SAM has two main contributions. On the one hand, SAM can suppress useless information while emphasizing useful information, which can also pass important spatial information and spectral information to the next stage, enhancing the network's ability to learn features. On the other hand, SAM can monitor the recovered fused image and calculate the loss with the ground truth to give full play to the performance of the network.

As described in Fig. 4, the feature map with the spatial dimension of $W \times H$ and the number of channels C is used as the input of the SAM. First, img_1 is obtained by 1×1 convolution, and then, img_1 is added to the input image img_MP to obtain the restored image img_RS. As mentioned earlier, we provide supervision on the feature map img_RS to calculate the loss between img_RS and ground truth. Then, the pixel attention mask img_M is generated from img_RS through a 1×1 convolution and the activation function sigmoid. After that, img_M is used to recalibrate the convolutionally transformed img_MP to obtain the attention guidance feature, which is added to the entity mapping path. Finally, a feature map with a spatial dimension of $W \times H$ and a channel number of C is obtained as the attention-enhancing feature map.

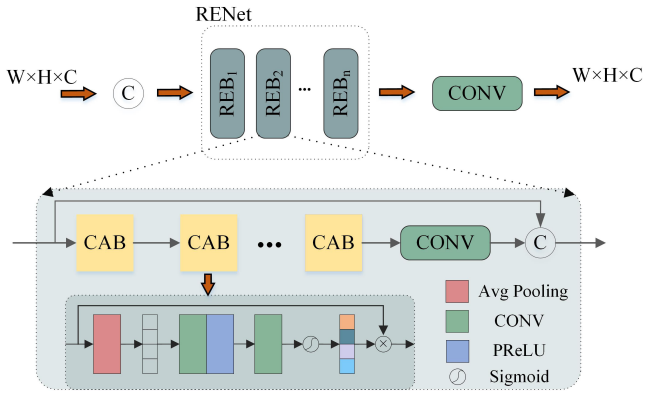

Fig. 5. Detailed flowchart of the RENet. RENet consists of REBs that nest the CAMs and the convolution layers.

4) Resolution Enhancement Net: To make the final generated fused result closer to the HRMS image, we added an RENet in Stage 3, which is shown in Fig. 5. It can be seen that RENet is composed of many resolution enhancement blocks (REBs), among which the CAM and convolution are nested in the REB to generate high-resolution features. The spatial dimension and the number of channels of the feature map remain unchanged throughout the RENet.

C. Loss Function

Unlike other methods, our method calculates the loss by combining the image at each stage with the ground truth. Given any Stage, add img_1 to img_MP to get the restored image img_RS as mentioned earlier in the SAM. Therefore, we will optimize MRFNet using the following loss:

$$\text{LOSS} = \sum_{S=1}^{3} [L_C(X, Y) + \lambda L_E(X, Y)] \tag{3}$$

where X represents the restored image img_RS and Y represents the ground truth. λ controls the relative importance of the two loss functions, which is set to 0.05 here. L_C stands for Charbonnier loss [56], which is a variant of loss. It can be expressed as

$$L_C = \sqrt{\|X - Y\|^2 + \varepsilon^2} \tag{4}$$

where ε is set to 10^{-3} for ensuring the stability of the value in the experiment. Moreover, L_E represents edge loss, which is defined as

$$L_E = \sqrt{\|\Delta(X) - \Delta(Y)\|^2 + \varepsilon^2} \tag{5}$$

where Δ represents the Laplace operator.

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TABLE I
SPECTRAL WAVELENGTH AND SPATIAL RESOLUTION OF WORLDVIEW II, GF-2, AND QB

Satellites	Spectral Bands (nm)					Spatial Resoution (m)	
	Blue	Green	Red	Nir	PAN	PAN	MS
WorldView II	450-510	510-580	630-690	770-895	450-1040	0.5	1.8
GF-2	450-520	520-590	630-690	770-890	450-900	1	4
QuickBird	450-520	520-660	630-690	760-900	450-900	0.61	2.44

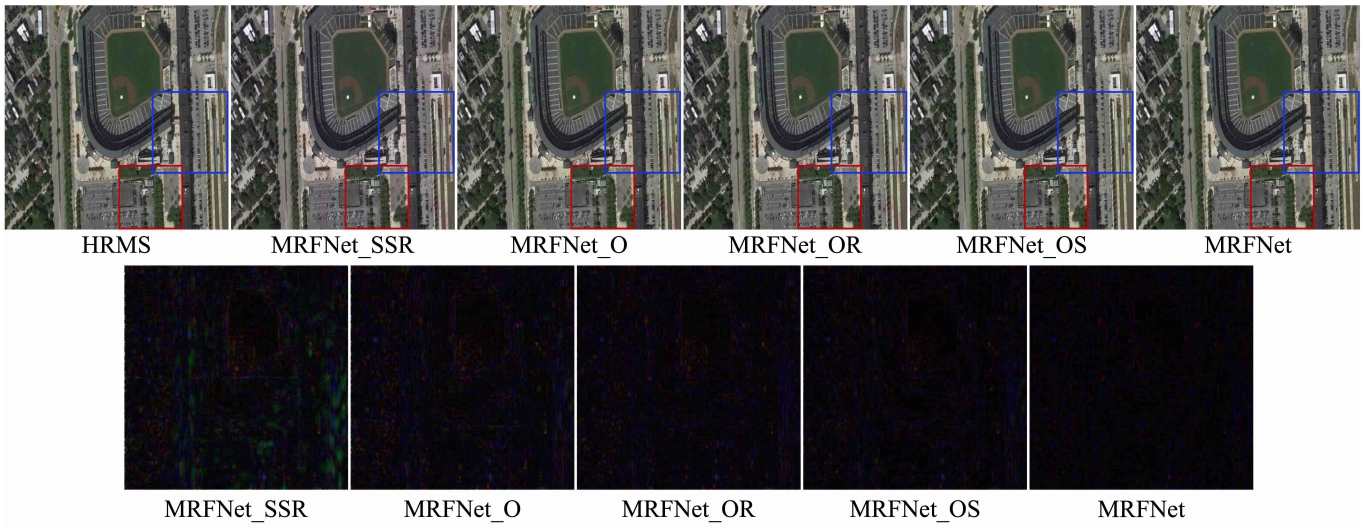

Fig. 6. Results of five different models on the dataset GF-2. The first line is a set of fused results, and the second line is a set of residual images between results and the HRMS image, respectively, which can be zoomed in for details.

IV. EXPERIMENT AND EVALUATIONS

In this section, we first introduce the datasets and evaluation indicators used in the experiment, and then, the ablation experiment is executed. Finally, the proposed method is compared with seven state-of-the-art pansharpening methods to prove the effectiveness of our model. Our experiments were performed on the NVIDIA Titan RTX computer using Pytorch. In addition, we performed two types of comparative experiments: experiments with reference and nonreference experiments, each of which includes qualitative visual effects analysis and quantitative evaluation.

A. Dataset and Metrics

We have evaluated the proposed MFRNet on three benchmark datasets (see Table I), including WorldView II (WV2), GF-2, and QuickBird (QB). Both three satellites contain four bands of red, green, blue, and near-infrared. WV2 can provide the PAN image with a spatial resolution of 0.5 m and the MS image with 1.8 m, and GF-2 can provide the PAN image with a spatial resolution of 1 m and the MS image with 4 m, while QB can provide the PAN image with a spatial resolution of 0.61 m and the MS image with a spectral resolution of 2.44 m. We use two main methods to evaluate the experimental results, respectively, namely, qualitative analysis and quantitative evaluation. On the one hand, we visualize the image so that it can be clearly displayed on the monitor in the qualitative analysis. On the other hand, we use several common indicators to objectively evaluate the fused results in the quantitative evaluation, which are divided into evaluation indicators with reference and nonreference evaluation indicators. Reference evaluation indicators include Erreur Relative Globale Adimensionnelle de Synthese (ERGAS) [57], spectral angle mapper (SAM) [57], [58], spatial correlation coefficient (SCC) [58], and image quality (Q). The introduction of evaluation indicators with reference is defined as follows.

1) ERGAS: It is used to calculate the global distortion, which is calculated by RMSE [59]. The value of ERGAS is smaller, indicating that the fusion effect is better.

2) SAM: It takes each spectral band as a coordinate axis and then calculates the angle between the fused result and the reference image corresponding to the pixels on the spatial coordinate axis. The smaller the value of SAM, the less spectral distortion, and the ideal value of which is 0.

3) SCC: It represents the similarity of spatial feature between the fusion image and the reference image. SCC takes a value between $[-1, 1]$. The higher the value, the better the preservation of the spatial information of the fused result, and the higher the image quality.

4) Q: As an overall evaluation index, Q measures the similarity of spatial information, brightness, and contrast between the fused result and the reference image. Similarly, the closer the value of Q is to 1, the better the quality of the fused result.

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TABLE II

FIVE DIFFERENT MODELS ARE PROPOSED FOR ABLATION STUDY. STAGE REFERS TO THE NUMBER OF MODEL'S STAGES, AND U-NET PRESENTS THE NUMBERS OF SETS OF ENCODER–DECODER

Network	Stage	U-Net	SAM	RENet
MRFNet_SSR	2	1	✓	✓
MRFNet_O	3	2	-	-
MRFNet_OS	3	2	✓	-
MRFNet_OR	3	2	-	✓
MRFNet	3	2	✓	✓

In addition, for the real data without reference image, we use three nonreference evaluation indicators, including D_λ, D_S, and QNR [60]. D_λ represents the degree of spectral information distortion, and D_S represents the degree of spatial information distortion. QNR is used to measure the distortion degree of spatial information and spectral information between the real data fused result, and the original input image, which is calculated by D_λ and D_S, can be defined as

$$\text{QNR} = (1 - D_S)^\alpha (1 - D_\lambda)^\beta. \tag{6}$$

B. Ablation Study

To verify the effectiveness of each module in the proposed network, we conducted ablation experiments on the dataset GF-2. Five different models are designed for comparison (see Table II) to verify the effects of each part of the proposed MRFNet:

1) MRFNet_SSR: Compared with the proposed MRFNet, MRFNet_SSR only uses a set of encoder–decoder for feature extraction, and the second stage is the same as the third stage of MRFNet (with SAM and RENet).

2) MRFNet_O: As the original network architecture, MRFNet_O maintains the same three-stage process as MRFNet, which uses two sets of codecs to extract features and, finally, uses concat for fusion (no SAM and RENet).

3) MRFNet_OR: MRFNet_OR adds RENet to the third stage on the basis of model MRFNet_O to maintain the spectral information and spatial information of the fused result (no SAM).

4) MRFNet_OS: MRFNet_OS adds the SAM to the proposed three-stage MRFNet_O to strengthen the connection between every two stages (no RENet).

5) MRFNet: MRFNet is our proposed network, which adds RENet to the third stage on the basis of MRFNet_OS to maintain the spectral information and spatial information of the fused result (with SAM and RENet).

Fig. 6 shows the fused results and residual images of the five types of models. Figs. 7 and 8 are enlarged views of the results, respectively. In Fig. 6, the fused result of MRFNet_SSR has a slight deviation in color distribution compared with other images. From the residual images, the spectral distribution of MRFNet_SSR is disordered. Compared with MRFNet_OR, MRFNet_O has similar spectral distributions with that in MRFNet_OR, but some details of the MRFNet_OR fusion image are more complete than MRFNet_O. MRFNet_OS adds the SAM module, which increases the spectral resolution of the fused result significantly. However, our proposed network has a uniform color distribution while retaining the integrity of the spatial information.

Fig. 7. Enlarged details of the fused results in Fig. 6 (red box).

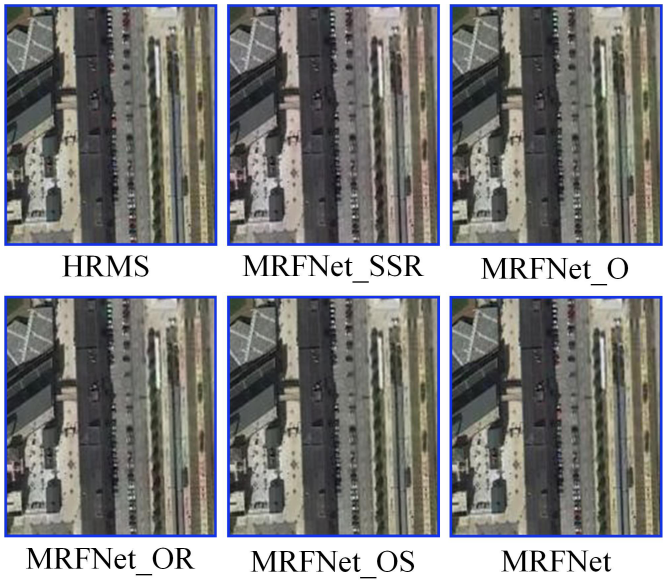

Fig. 8. Enlarged details of the fused results in Fig. 6 (blue box).

Fig. 7 shows some details of the fused results of each module. It is obvious that, in the result of MRFNet_SSR, the roof and gray roads have red deviations, and the result cannot show the original colors of the trees basically. The result of MRFNet_O has been improved, but the colors of the roof and the trees are still deviated and the house next to the trees is greenish. The problem of red deviations in MRFNet_OR has disappeared, but the houses next to the trees are still greenish. MRFNet_OS improves the greenishness of houses, but the trees are darker than the HRMS image in color. Furthermore, MRFNet can display the information of ground features and maintain a good spectral distribution.

The fused result of MRFNet_SSR in Fig. 8 has uneven and distorted color distribution, accompanied by the situation redness. In the fused result of MRFNet_O, the color distribution

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of the road on the right is uneven, and the information of vehicles staying on the road cannot be displayed. The above color distortion problem still exists in the result of MRFNet_OR, but it can display part of the vehicle information. The fused result of MRFNet_OS does not show the phenomenon of color disturbance in the road, but its spatial information is lost. However, MRFNet can fully display the vehicle information of the parking garage and retain the complete spectral distribution relatively. In conclusion, the proposed method shows better performance, which proves the effectiveness of the proposed MRFNet.

C. Comparison With SOTAs

In this part, we compare the proposed method with the advanced method on simulated data and real data, including qualitative analysis and quantitative comparison. Furthermore, the traditional methods consist of P + XS [27], GSA [2], IHS [13], and generalized LP with MTF-matched filter and multiplicative injection model (MTF_GLP_HPM) [28]. Deep-learning-based methods include PanNet [36], PNN+ [35], and PanGAN [45]. The codes of all compared methods are public, and the parameters are set according to the original papers.

1) Experiments on Simulated Data: According to Wald's protocol, we have performed experiments on the simulated data, where the HRMS image is used as the ground truth to play a role of the reference image, including qualitative and quantitative experiments. Qualitative analysis was conducted at first. Fig. 9 shows the experimental results on simulated data from benchmark datasets WV2, GF-2, and QB in eight methods. It can be seen that all the methods can recover the fused image, but there are some problems of spatial detail loss and spectral distortion. It is obvious that the P + XS method has the problem of fuzzy spatial information and cannot present useful information of ground objects in the eight methods. In, addition, on Img1 and Img3, the spectral information retained by P + XS is more complete, while the spectral information lost by PNN+ is serious. In the first set of images, the rivers of the seven compared methods have different degrees of red blocks, but this phenomenon does not appear in the results of the proposed method. For Img2, PNN+ has severe spectral distortion. Meanwhile, in addition to PNN+ and our method, the remaining six methods have better spectral information retention capabilities but also have more or less spatial information distortion problems. Moreover, our MRFNet retains as much spatial information as possible while also maintaining a more uniform spectral distribution. For Img3, all methods except PNN+ have rich spectral information, but the white roof in MTF_GLP_HPM and PanNet has overexposed and lost spatial details. In Img4, the spectral distribution in the fused result of MTF_GLP_HPM is relatively uniform, but there is obvious distortion in the upper left corner. In the results of PNN+ and P + XS, the loss of spatial information has occurred, and P + XS is more serious. In contrast, the other several methods can better preserve spatial information. However, compared with the ground truth, the spectral information is somewhat lost. In Img5, the green of the image becomes lighter, and the color patches of the seawater are unevenly distributed in the compared method. In Img6, the color of seawater should have been dark green. For this, the proposed method retains better conditions. However, in the other seven methods, the color of seawater becomes dark blue. It can be seen that the spatial information of the PNN+ and P + XS methods is incomplete, and there is a fuzzy phenomenon.

Figs. 10 and 11 are detailed images of the eight methods on the simulated data of GF-2 and QB, in which the upper part is the fused result and the lower part is the magnified detail image. In Fig. 10, for the runway portion of an airport, P + XS reveals little useful information. Compared with P + XS, PNN+ can show the general outline of the runway, while the color blocks of MTF_GLP_HPM and PanNet are unevenly distributed. GSA, IHS, PanGAN, and our proposed method can preserve spatial information and spectral information well. The spatial details of P + XS and PNN+ are blurred in the enlarged images of aircraft details. As the result of the PanNet method, the edges of the aircraft are blurry, and the spectral information is lost to a certain extent. In the other methods and ours, the shape of the airplane remains relatively intact, and the spectral distribution is uniform. In Fig. 11, other methods have distortions while maintaining the white part of the spectrum in the red frame except for our method. More serious is that P + XS can hardly display useful information. For the detailed zoom part in the blue box, in addition to P + XS, the other six compared methods and our method can maintain the spatial information and spectral information better.

Figs. 12, 13, and 14 measure the residual images [45] of the eight methods on GF-2, QB, and WV2, respectively, which refers to the difference between the fused results of each method and the ground truth. In Fig. 12, it can be seen that the spectral distortion of PNN+ and the spatial information loss of P + XS are the most serious. It is obvious that PanGAN has a deviation in the green grass part, and PanNet has a slight deviation in the red runway part. Compared with GSA and IHS, the spectral distortion and spatial information loss of MTF_GLP_HPM are more serious. Our method is close to the ground truth and has the smallest error, which can be zoomed in to view; there is a small amount of spectral distortion and negligible loss of spatial details in the residual image of the proposed method. Fig. 13 shows the fused results and residual images of eight methods on the dataset QB. From the first row of fused results, each method can better learn the spectral information in the MS image, where the spectral distribution is relatively uniform. The residual images in the second row can more intuitively represent the difference in detail between the fused results and the reference image. It can be seen that the color of the road in the result of PNN+ is light red. This kind of regional spectral distortion also occurs in MTF_GLP_HPM. For the loss of spatial information, P + XS is the most serious, and the details of IHS, PanNet, and PanGAN are less lost. Zoom in to observe that our MRFNet performs well regardless of the spectral information or spatial details. In Fig. 14, P + XS has the largest loss of spatial information, followed by PNN+ and PanNet. In terms of spectrum preservation, MTF_GLP_HPM has obvious spectral distortion, and the red area appears on the shore of the fused

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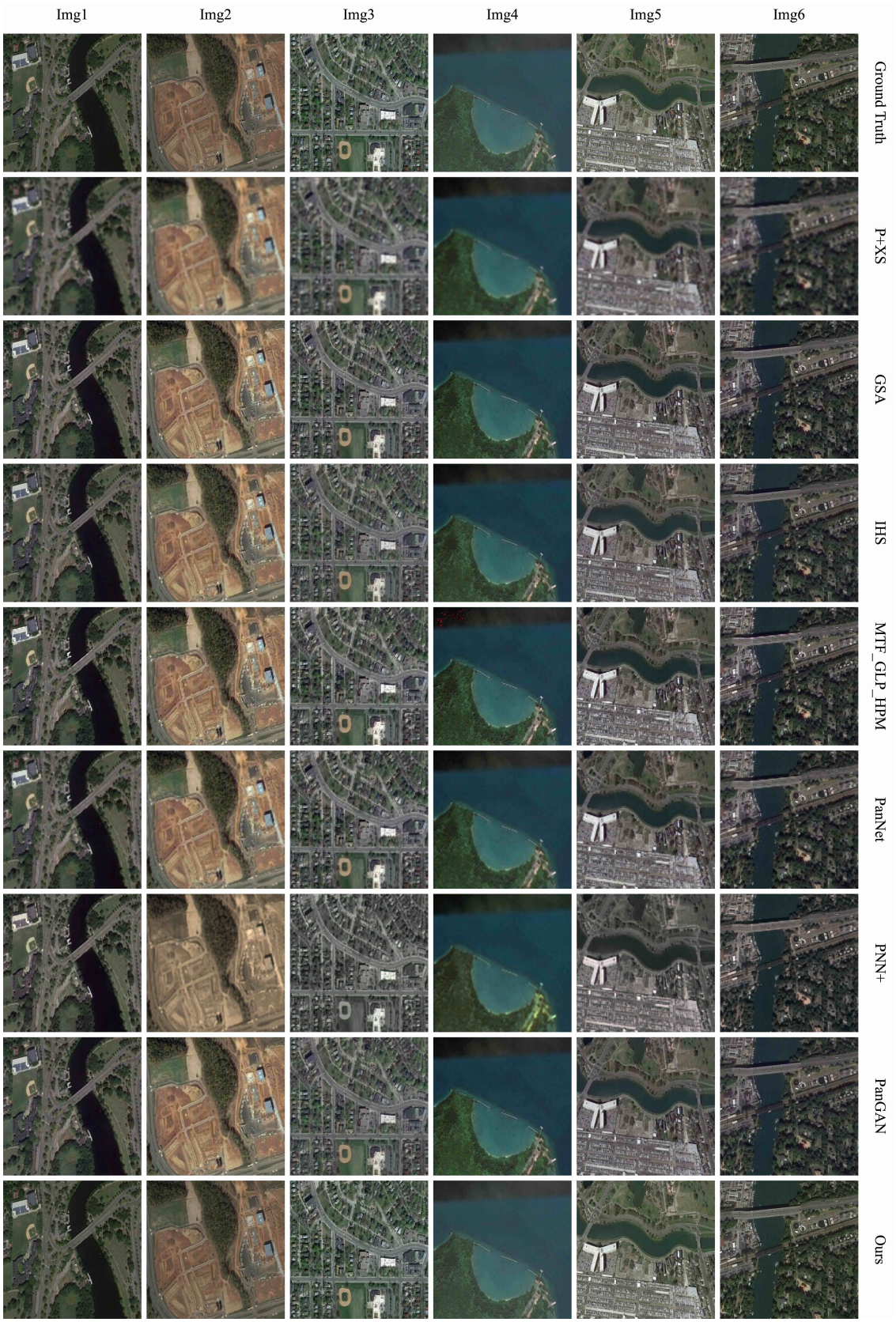

Fig. 9. Experimental results of eight pansharpening methods on simulated datasets from WV2, GF-2, and QB. The first row is the ground truth, and the second to the ninth rows are experimental results, which can be zoomed in to view details.

result. The color of seawater in PanGAN and ground truth has some errors, and the spectrum is unevenly distributed in the ship part of the GSA fused result. Enlarging our fusion results, we can see that there are spectral distortion problems in the houses on the shore, but, compared to other methods, our method has less error and is closest to the ground truth.

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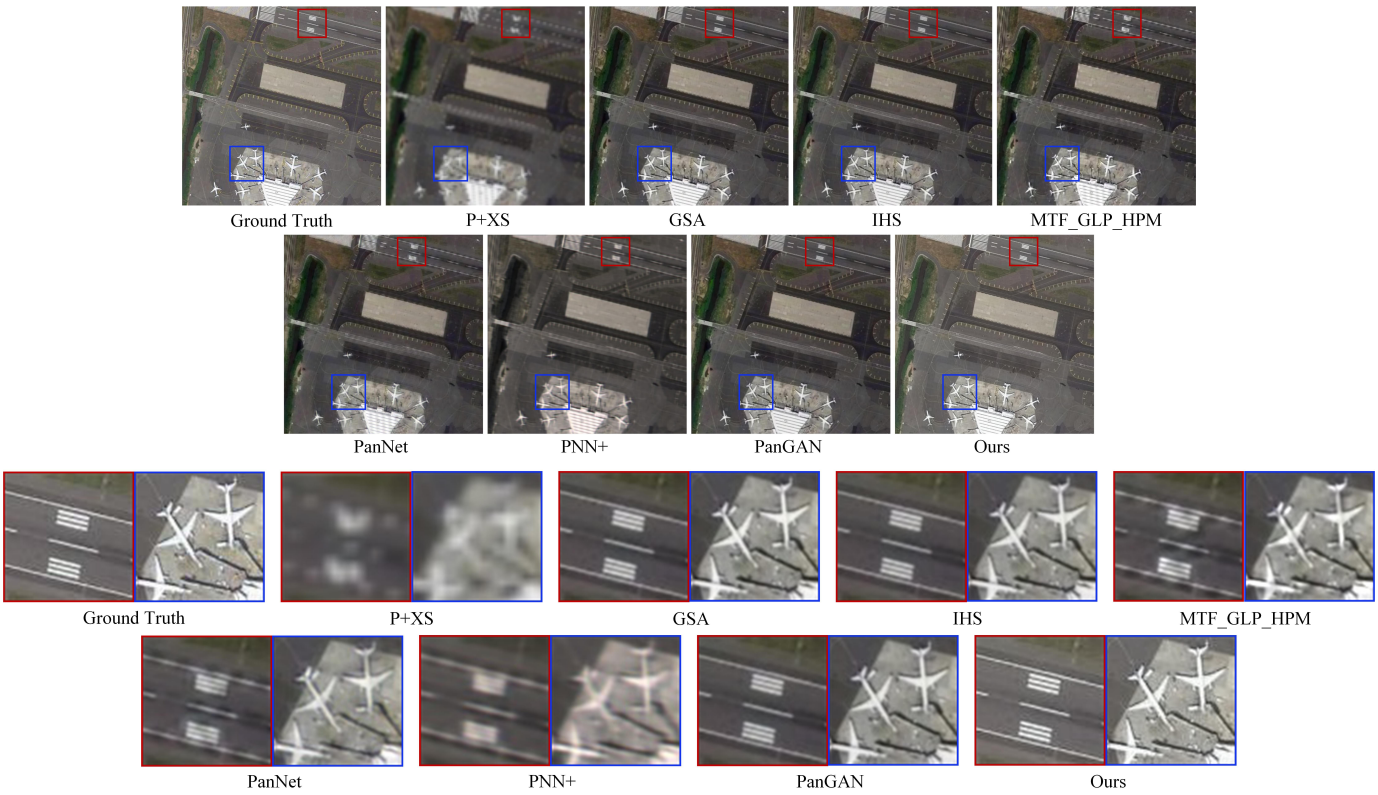

Fig. 10. Results of different pansharpening methods on the simulated dataset from GF-2. The upper part shows the fused results of different methods, and the lower part represents the enlarged details of the fused result (red and blue frames).

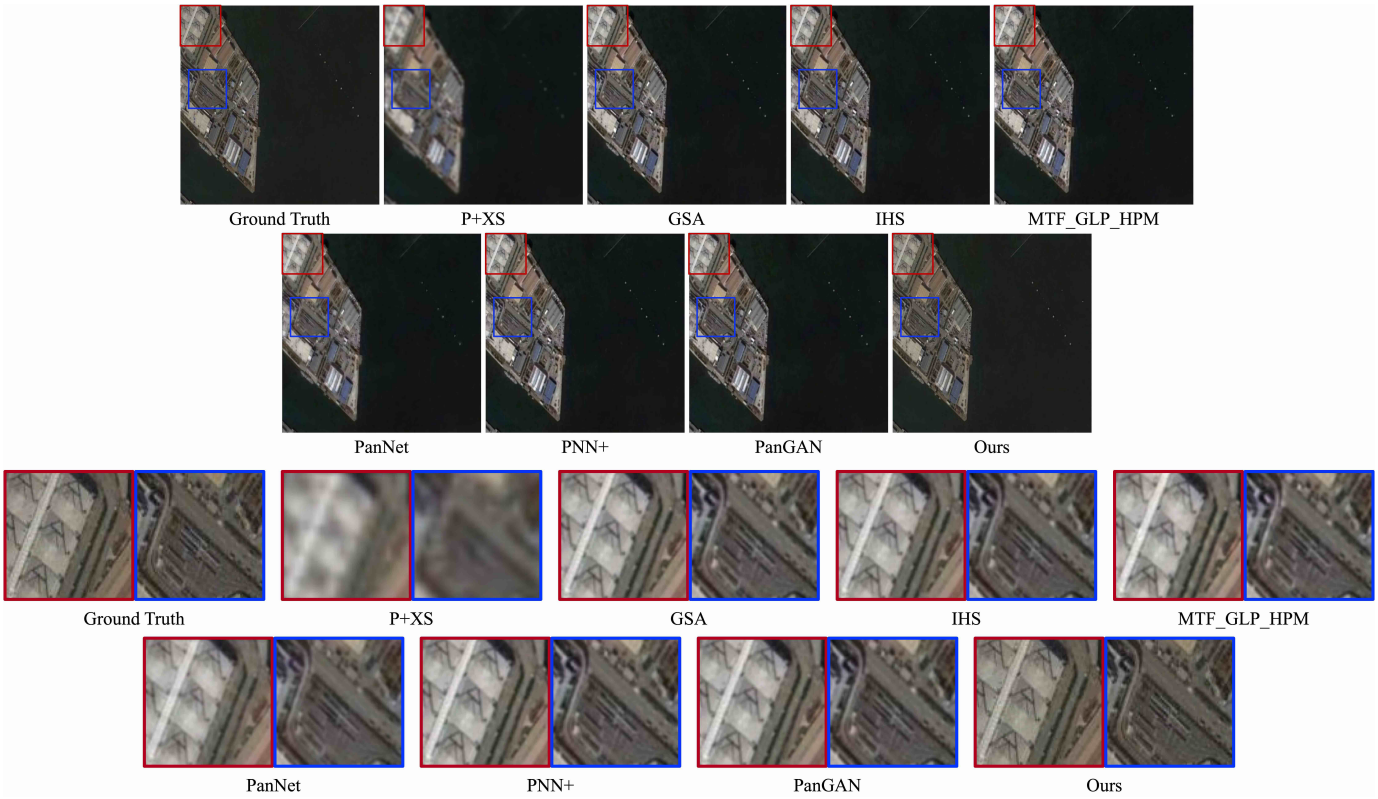

Fig. 11. Results of different pansharpening methods on the simulated dataset from QB. The upper part shows the fused results of different methods, and the lower part represents the enlarged details of the fused result (red and blue frames).

In addition, qualitative comparison on three datasets is performed, respectively. The mean values of the evaluation indexes of the fused images on WV2, GF-2, and QB are shown in Tables III–V. It can be seen that we used four evaluation indicators with reference for the experiment, including ERGAS, SAM, SCC, and Q. Whether the simulated data

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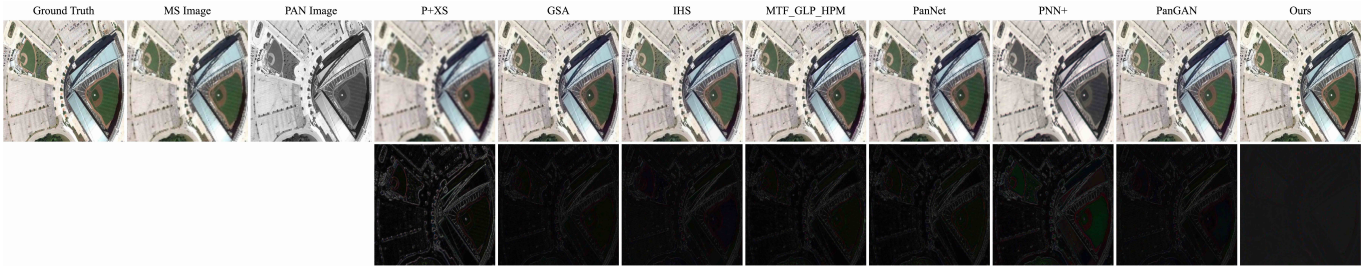

Fig. 12. Results of different pansharpening methods on the simulated dataset from GF-2 and their residual images with the ground truth. The first three columns are ground truth, MS image, and PAN image. The rest are the results and the corresponding residual images, which can be zoomed in to view the details.

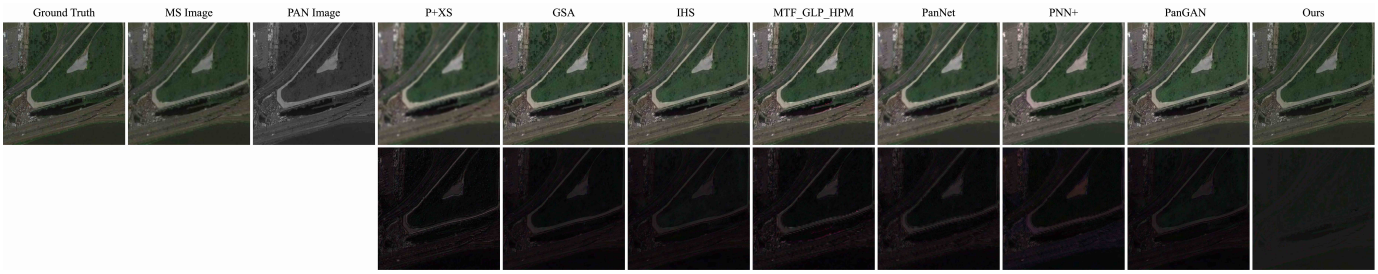

Fig. 13. Results of different pansharpening methods on the simulated dataset from QB and their residual images with the ground truth. The first three columns are ground truth, MS image, and PAN image. The rests are the results and the corresponding residual images, which can be zoomed in to view the details.

TABLE III
QUANTITATIVE EVALUATION OF FUSED RESULTS ON SIMULATED DATASET FROM WV2

Method	Worldview II			
	ERGAS ↓	SAM ↓	SCC ↑	Q ↑
P+XS	11.2499	0.1797	0.8860	0.8339
GSA	6.6593	0.1604	0.9452	0.8877
IHS	5.3839	0.1531	0.9375	0.8912
MTF_GLP_HPM	6.1424	0.1692	0.9348	0.8605
PanNet	4.9368	0.1329	0.9564	0.9511
PNN+	5.9874	0.1526	0.9228	0.9303
PanGAN	4.3101	0.1229	0.9400	0.9499
MRFNet	**2.1361**	**0.0364**	**0.9865**	**0.9665**

Bold indicates the best result.

TABLE IV
QUANTITATIVE EVALUATION OF FUSED RESULTS ON SIMULATED DATASET FROM GF-2

Method	GF-2			
	ERGAS ↓	SAM ↓	SCC ↑	Q ↑
P+XS	7.4854	0.1967	0.6894	0.7325
GSA	6.0122	0.1878	0.8459	0.8530
IHS	6.0982	0.1753	0.8707	0.8734
MTF_GLP_HPM	6.1477	0.1866	0.8380	0.8201
PanNet	4.2780	0.1078	0.9002	0.8730
PNN+	4.4430	0.1537	0.8838	0.8552
PanGAN	3.0982	0.1246	0.9279	0.8765
MRFNet	**1.9139**	**0.0562**	**0.9749**	**0.9277**

Bold indicates the best result.

TABLE V
QUANTITATIVE EVALUATION OF FUSED RESULTS ON SIMULATED DATASET FROM QB

Method	QuickBird			
	ERGAS ↓	SAM ↓	SCC ↑	Q ↑
P+XS	5.1504	0.1799	0.7957	0.6351
GSA	3.9494	0.1538	0.9500	0.8174
IHS	4.9835	0.1888	0.9399	0.7833
MTF_GLP_HPM	4.9191	0.1783	0.9455	0.8042
PanNet	2.9775	0.0922	0.9585	0.8632
PNN+	3.1835	0.1344	0.9351	0.8490
PanGAN	2.8202	0.1006	0.9587	0.8679
MRFNet	**1.5669**	**0.0246**	**0.9769**	**0.9022**

Bold indicates the best result.

come from WV2, GF-2, or QB, the values of indicators on the proposed method are better than those on traditional methods and deep-learning-based methods. Table III is the comparative experiment conducted on WV2. Compared with PanGAN, our SAM value is reduced by 0.0865, which shows that our result retains more complete spectral information. Compared with PanNet, our SCC value is improved by 0.0301, indicating that our spatial information is richer. Table IV is an experiment conducted on GF-2. Compared with PanNet, our SAM value is reduced by 0.0516, which proves that our multilevel network can preserve the spectrum more completely. Compared with PanGAN, our SCC value has increased by 0.047, indicating that the addition of each module can help us restore image details. Table V shows the quantitative comparison of each method on QB. It is obvious that the value of EGRAS value in our method is decreased by 1.2533 compared with PanGAN, and the value of SAM is decreased by 0.0676 compared with PanNet, which indicates that our MRFNet could retain spatial information and spectral information effectively.

2) Experiments on Real Data: In addition to simulated experiments with reference images, we conducted tests and compared experiments on real datasets from the benchmark datasets WV2, GF-2, and QB, which does not have the ground truth as the reference image. Therefore, three nonreferenced evaluation indicators are used to conduct quantitative experiments on the three types of datasets. Figs. 15 and 16 show the fused results of different pansharpening methods on the

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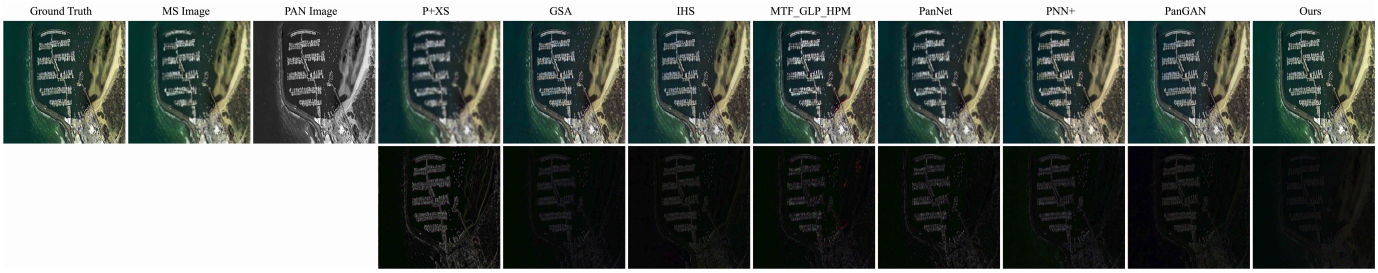

Fig. 14. Results of different pansharpening methods on the simulated dataset from WV2 and their residual images with the ground truth. The first three columns are ground truth, MS image, and PAN image. The rests are the results and the corresponding residual images, which can be zoomed in to view the details.

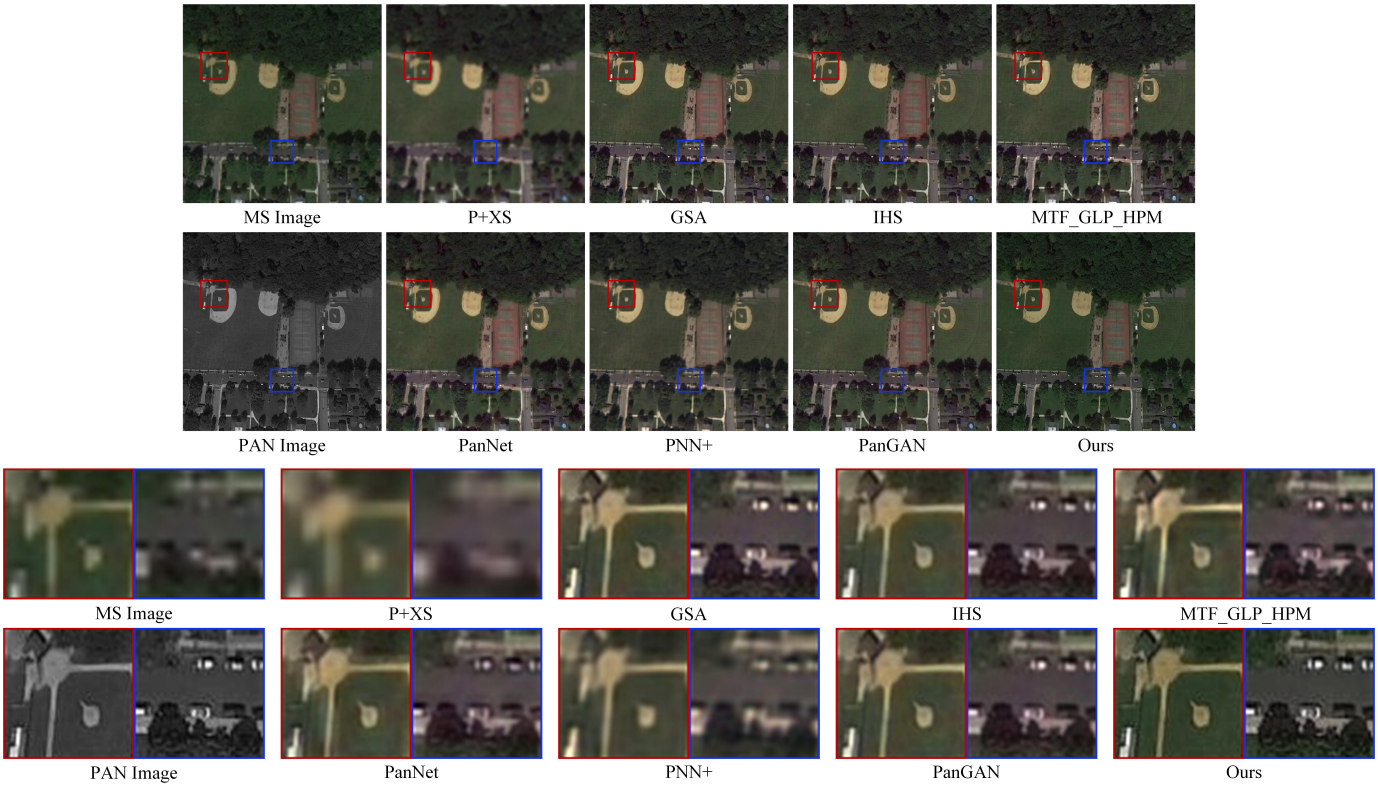

Fig. 15. Fused results of different pansharpening methods on a real dataset from WV2. The upper part shows the fusion results of different methods, and the lower part represents detailed enlargements of the fusion results (red and blue frames), respectively.

real dataset from WV2 and QB, respectively. The upper part represents the results of each method, and the lower part shows the detailed enlarged images (red and blue frames), respectively. It can be seen from the upper part of the results that the other methods do not have much spectral distortion and spatial detail blurring except for the fuzzy results of P + XS. The lower part of the display is clearer. It can be seen that the spatial details of the MS image are relatively fuzzy, but the spectrum is rich, and the spatial details of the PAN image are relatively clear but lack spectral information. The result of P + XS has the most obscure spatial details but retains spectral information. For the enlargement of the lawn, the fused result of IHS and PNN+ retains information in the yellow circle incompletely, and the edge is illusory. Furthermore, except for MTF_GLP_HPM, the spectral information of each method is better preserved, and there is no serious spectral distortion. For the second enlarged image of road vehicles, all other methods showed redness except for our method and PNN+, which is inconsistent with the spectrum displayed by the MS image. In terms of spatial information retention, PNN+ has better detail retention than P + XS, which does not show any useful information basically. Apart from them, other compared methods and ours can retain good spatial information. In Fig. 16, the spectral information distribution of the eight methods is relatively uniform visually, but the retention of spectral details is different. In the enlarged details of the first image showing the land (red frame), P + XS cannot show the ground feature information at all. Except for our method, the colors of the other six methods are darker than the MS image, and the spectral information retained by our results is basically the same as that of the MS image. For the second enlarged part (blue frame), our method has

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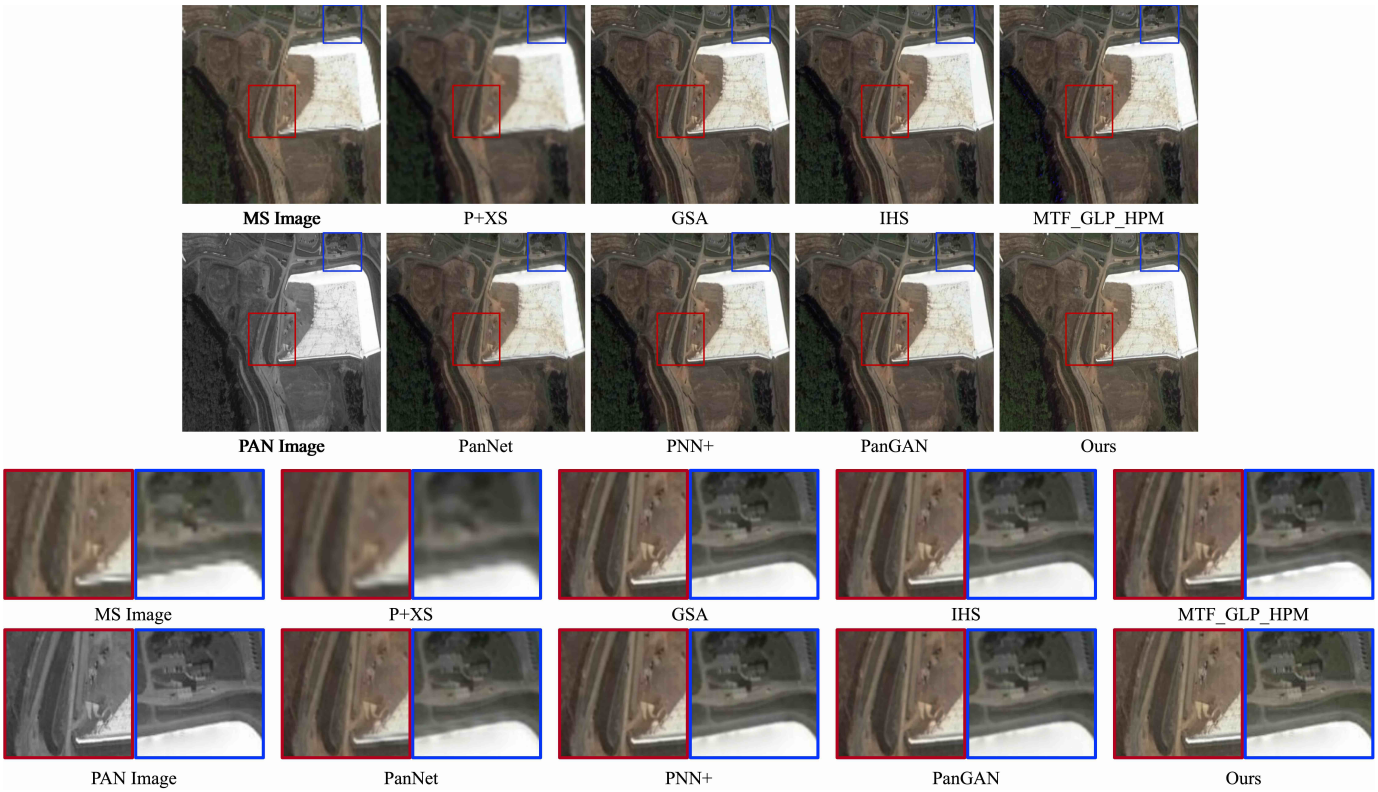

Fig. 16. Fused results of different pansharpening methods on a real dataset from QB. The upper part shows the fusion results of different methods, and the lower part represents detailed enlargements of the fusion results (red and blue frames), respectively.

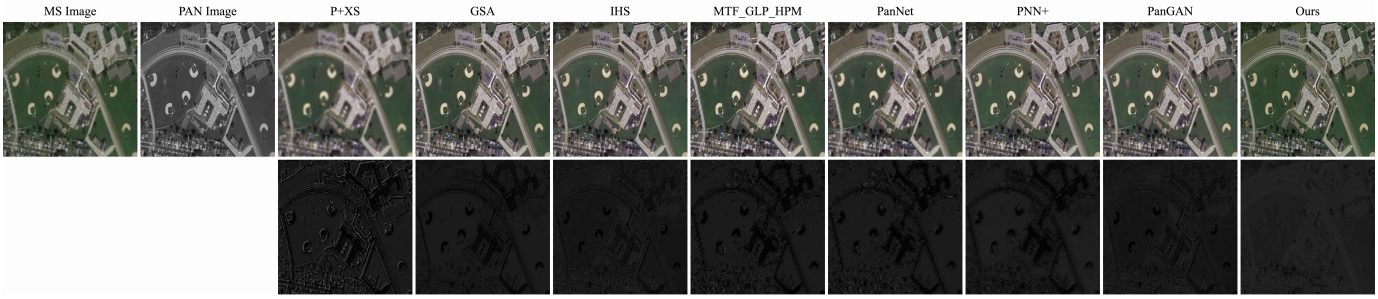

Fig. 17. Fused results of different pansharpening methods on real datasets from WV2 and gradient residual images of fused results with the PAN image. The first two columns are MS images and PAN images, respectively, and the rest are fused results and corresponding gradient residual images, which can be zoomed in for details.

a better ability to retain spatial information. After zooming in, we can see the outline information of the vehicle and the house in the upper right corner clearly. In traditional methods, the spectral distribution is disordered, which is inconsistent with the description of the MS image.

Figs. 17, 18, and 19 are the fused results of different pansharpening methods on WV2, QB, and GF-2 and their gradient residual images with the PAN image because there is no ground truth as the reference. In Fig. 17, it is obvious that the fused result of P + XS is the most blurred, which is far from the details of the PAN image. Correspondingly, the error image is the most obvious, which is consistent with the visual effect. Among the three methods of GSA, his, and MTF_GLP_HPM, the detailed information of the result in IHS is the closest to the PAN image, that is, the spatial information is maintained better. Among the three deep learning methods of PanNet, PNN+, and PanGAN, the spatial details of the fused result in PanGAN are relatively clear. Furthermore, the fused result of the proposed method is close to the MS image and the spatial information close to the PAN image, which is better than other methods. In Fig. 18, there is no significant deviation in the spectral information of the eight pansharpening methods visually. When the gradient error images corresponding to the fusion results are magnified, the spatial information loss of P + XS is the most serious. In the set of results, the spectral distribution of the fusion results of P + XS, MTF_GLP_HPM, and PanNet is somewhat chaotic, while the spectral distribution of the fusion results of other methods is more uniform. For the retention of spatial information, PanGAN, IHS, and our methods performed well. Similar conclusions can be drawn when analyzing Fig. 19. It is obvious that the result of P + XS is more ambiguous, that is, the spatial information

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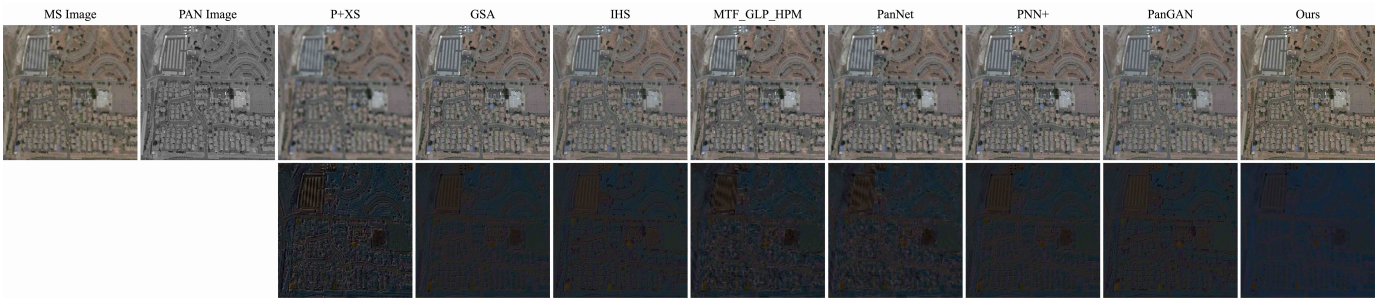

Fig. 18. Fused results of different pansharpening methods on real datasets from QB and gradient residual images of fused results with the PAN image. The first two columns are MS images and PAN images, respectively, and the rest are fused results and corresponding gradient residual images, which can be zoomed in for details.

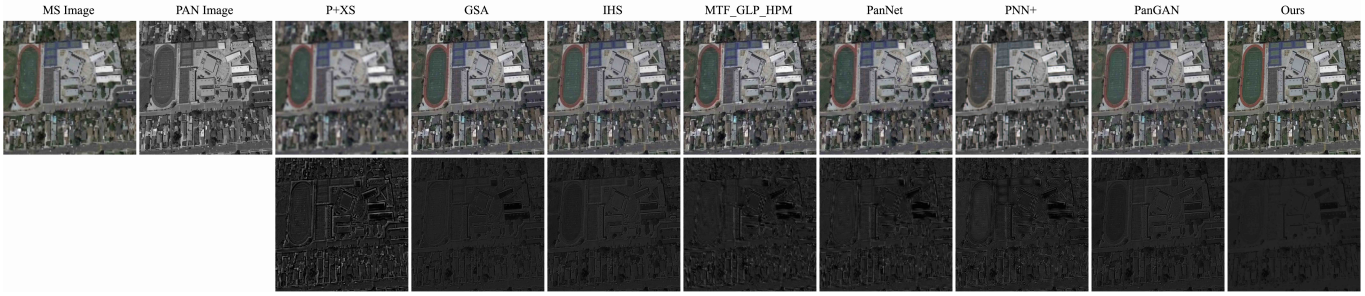

Fig. 19. Fused results of different pansharpening methods on real datasets from GF-2 and gradient residual images of fused results with the PAN image. The first two columns are MS images and PAN images, respectively, and the rest are fused results and corresponding gradient residual images, which can be zoomed in for details.

TABLE VI
QUANTITATIVE EVALUATION OF FUSED RESULTS ON REAL DATASETS FROM WV2, GF-2, AND QB

Method	WorldView II			GF-2			QB		
	QNR ↑	D_λ ↓	D_S ↓	QNR ↑	D_λ ↓	D_S ↓	QNR ↑	D_λ ↓	D_S ↓
P+XS	0.8295	0.0481	0.1286	0.7634	0.0278	0.2147	0.7015	0.0762	0.2406
GSA	0.9039	0.0280	0.0597	0.8958	0.0441	0.0628	0.8543	0.0778	0.0737
IHS	0.9399	0.0364	0.0246	0.8925	0.0573	0.0533	0.8671	0.0703	0.0674
MTF_GLP_HPM	0.8971	0.0374	0.0681	0.8456	0.0989	0.0615	0.8676	0.0594	0.0777
PanNet	0.9527	0.0222	0.0257	0.9187	0.0609	0.0217	0.8821	0.0622	0.0594
PNN+	0.9342	0.0175	0.0491	0.9023	0.0796	0.0196	0.8701	0.1103	0.0220
PanGAN	0.9520	0.0104	0.0380	0.9221	0.0435	0.0379	0.8743	0.0680	0.0726
MRFNet	0.9708	0.0096	0.0198	0.9437	0.0191	0.0360	0.9113	0.0387	0.0424

is seriously lost. In addition, the spectral distribution of the result in PNN+ is not uniform, which leads to the phenomenon of color spots and agglomeration in the image. Observing the gradient residual images in the second row, it can be seen that, compared with IHS, GSA retains more complete details of the playground, and IHS has advantages in retaining house information. Both PanNet and PNN+ have uneven spectral distribution, while PanGAN has good spectral retention. Moreover, the spectrum and spatial information retentions are relatively complete in the result of our method.

Table VI shows the quantitative comparison of eight methods on three real datasets from WV2, GF-2, and QB (red, green, and blue represent the first, second, and third, respectively). It can be seen that three nonreference evaluation indicators are used, including D_λ, D_S, and QNR to calculate the degree of spectral information distortion, the degree of spatial detail loss, and the overall quality of fused image, respectively. As Table VI shows, on the left part of the WV2 dataset, the QNR value of our method is increased by 0.0181 compared to PanNet, which shows that our method has advantages in terms of spectral information retention. Besides, in terms of the value on D_S, our method reduces 0.0048 compared to IHS, indicating that the proposed method can better preserve spatial information. On the middle part of GF-2, our method increases the value of QNR by 0.0216 compared to PanGAN. In addition, in terms of the value on D_λ, the proposed method reduces 0.0087 compared to P + XS. For the indicators of real data from QB, the QNR value of our method is increased by 0.037 compared with that of PanNet, indicating that the overall fusion effect of the proposed MRFNet results is excellent. In the measurement of spectral distortion and spatial information losses by D_λ

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and D_S, our method also achieves better values. In general, whether qualitative analysis or quantitative comparison, our method can better integrate the spectral information of MS images with the spatial details of PAN images, which can obtain high-quality HRMS images.

V. Conclusion

In this article, we propose an MRFNet for MS images and PAN images pansharpening. The network consists of three stages, including two feature extraction stages and one image reconstruction stage, which are connected by SAMs and cross-stage fusion. In the feature extraction stages, we use the inverted residual block of network MobileNetV2 for subsampled and replace the skip connection between encoder and decoder with CAM. In the image reconstruction stage, RENet is added to generate the feature map, which is fused with the original fused image to maintain the spatial information and spectral information of the fused result. The ablation experiment proves the effectiveness of the proposed method. Furthermore, simulated experiments with HRMS images as reference images and real experiments without HRMS images are carried out on benchmark datasets WV2, GF-2, and QB, which are also compared with other methods. Experimental results demonstrate the superiority of the proposed method in visual subjective and objective evaluation indexes.

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